

Project: Incident Management Agent – AI-Assisted Root Cause Analysis Platform

Testing Framework: Pytest

Purpose

This document contains the primary happy-path test cases implemented to validate the core functionality of the Incident Management Agent application. The objective of these tests is to ensure that the main user workflows operate correctly under normal conditions.

Test Case 1 – User Authentication

Test Case ID

AUTH-001

Objective

Verify that a valid user can successfully log in to the system and receive an authentication token.

Preconditions

- User account exists in the system.
- Valid login credentials are available.

Test Steps

1. Navigate to the login page.
2. Enter valid email and password.
3. Submit the login request.

Expected Result

- User is authenticated successfully.
- JWT access token is generated.
- User is redirected to the dashboard.

Status

Pass

Test Case 2 – Create Incident

Test Case ID

INC-001

Objective

Verify that an authenticated user can create a new incident.

Preconditions

- User is logged into the system.

Test Steps

4. Navigate to the Create Incident page.
5. Enter incident title.

6. Enter incident description.
7. Provide incident timeline information.
8. Submit the incident.

Expected Result

- Incident is successfully created.
- Incident appears in the incident list.
- Incident details are stored in the database.

Status

Pass

Test Case 3 – Incident Category Classification**Test Case ID**

AI-001

Objective

Verify that the AI model correctly predicts an incident category.

Preconditions

- Incident data is available.
- Category classification model is loaded.

Test Steps

9. Create or open an incident.
10. Submit the incident for AI analysis.
11. Execute category prediction.

Expected Result

- System returns a valid category prediction.
- Predicted category is displayed in the incident details.

Status

Pass

Test Case 4 – Similar Incident Retrieval**Test Case ID**

AI-002

Objective

Verify that the FAISS similarity search retrieves related historical incidents.

Preconditions

- Historical incidents are available in the FAISS knowledge base.

Test Steps

12. Create or open an incident.

13. Trigger similarity search.
1. Review retrieved incidents.

Expected Result

- Similar incidents are returned when relevant matches exist.
- Retrieved incidents contain similarity scores and RCA references.

Status

Pass

Test Case 5 – Root Cause Analysis Generation**Test Case ID**

AI-003

Objective

Verify that the AI system generates a complete Root Cause Analysis report.

Preconditions

- Incident exists in the system.
- AI services are operational.

Test Steps

2. Open an incident.
3. Select "Generate RCA".
4. Wait for AI processing to complete.

Expected Result

- RCA report is generated successfully.

Report contains:

- Executive Summary
- Business Impact
- Timeline
- Evidence Collected
- Root Cause
- Contributing Factors
- Resolution Actions
- Preventive Measures

Status

Pass

Test Case 6 – RCA Document Download**Test Case ID**

RCA-001

Objective

Verify that generated RCA reports can be downloaded.

Preconditions

- RCA report has been generated.

Test Steps

5. Open generated RCA report.
6. Click Download Report.

Expected Result

- Report is downloaded successfully.
- Downloaded document contains complete RCA content.

Status

Pass

Test Case 7 – User Management

Test Case ID

ADMIN-001

Objective

Verify that an administrator can create and manage user accounts.

Preconditions

- Administrator account exists.
- Administrator is logged in.

Test Steps

7. Navigate to User Management.
8. Create a new user account.
9. Assign a role.
10. Save the user.

Expected Result

- User account is created successfully.
- User appears in the user management list.
- Assigned role is stored correctly.

Status

Pass